Internet of Things (IoT): Societal, Scientific, and Technical Foundations

An evidence-informed framework for secure, sustainable, and socially responsible connected systems

1 THE PROBLEM

Rapid infrastructure scaling

More connected devices increase system complexity, interoperability demands, and maintenance burden.

scale

Pervasive surveillance risks

Continuous sensing can expose behaviour, location, identity, and household data when governance is weak.

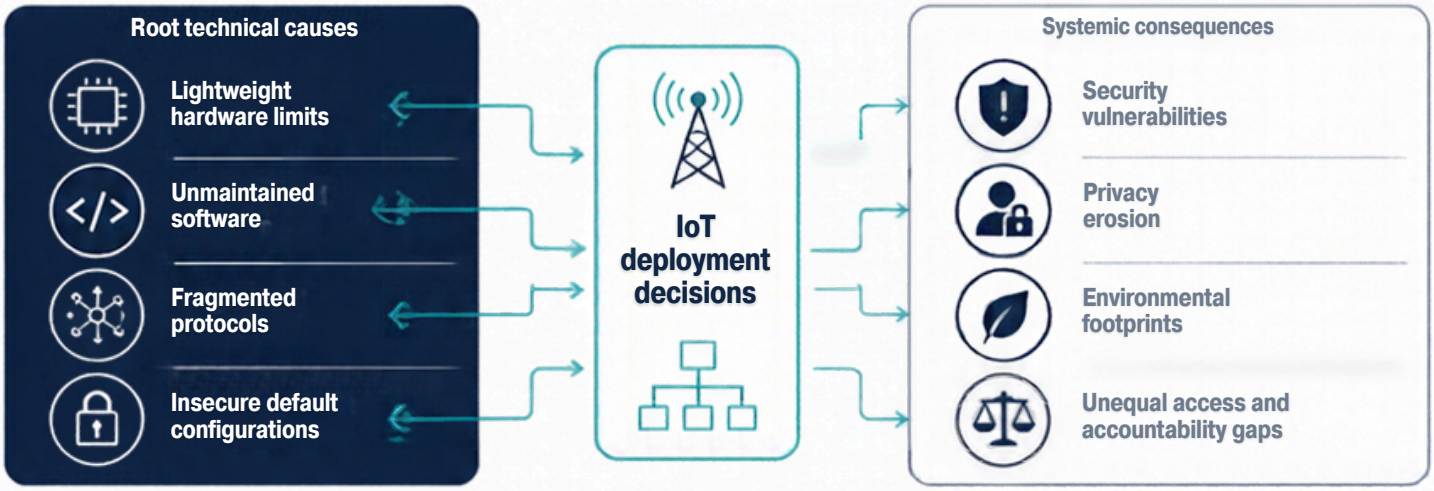
trust

Severe resource and energy constraints

Small embedded devices operate with limited power, memory, processing, and long service lifetiimes.

efficiency

2 CAUSES & CONSEQUENCES



Legend: Technical constraints can amplify societal and environmental impacts across the IoT lifecycle.

3 SUSTAINABLE & TECHNICAL SOLUTIONS

scalable by design

Edge computing

Process data closer to the source to reduce unnecessary transmission, latency, and centralisation.

Design Implication:
Reduce data movement; improve responsiveness; balance local resources and reliability.

Lightweight cryptography: Chacha20

Use efficient modern authenticated encryption to require a carefully selected cryptographic design.

Design Implication:
Lower compute and energy cost; strong confidentiality and integrity with minimal overhead.

Federaled Learning

Train models across distributed devices while keeping raw data local, with attention to long-lived leakage and update security.

Design Implication:
Preserve privacy; reduce data centralisation; secure aggregation and model updates.

Open-source embedded standards: RIOT OS

Support portable, auditable, developable and long-lived device ecosystems.

Design Implication:
Improve transparency; extend longevity; align with community-driven security and maintenance.

Security, privacy, energy, and usability must be evaluated together.

ACTION PLAN BY STAKEHOLDER

Governments / Regulators

Set baseline security requirements, require lifecycle support and transparent data governance, align procurement with sustainability.

Academic & Research Institutions

Evaluate real-world deployment effects; publish reproducable methods; teach secure systems, privacy, and responsive innovation.

Developers / Engineers

Treats-moded early, secure details, patchability: minm data collection inside data collection; measure energy and environmental track-offs

End-Users / Consumers

Review permissions; change default credentials; update devices; prefer transparent products; retire and recycle responsible.

KEY TAKEAWAY

Digital sovereignty begins with secure, sustainable IoT design.

Build connected systems that are technically resilient, socially accountable, and environmentally aware.